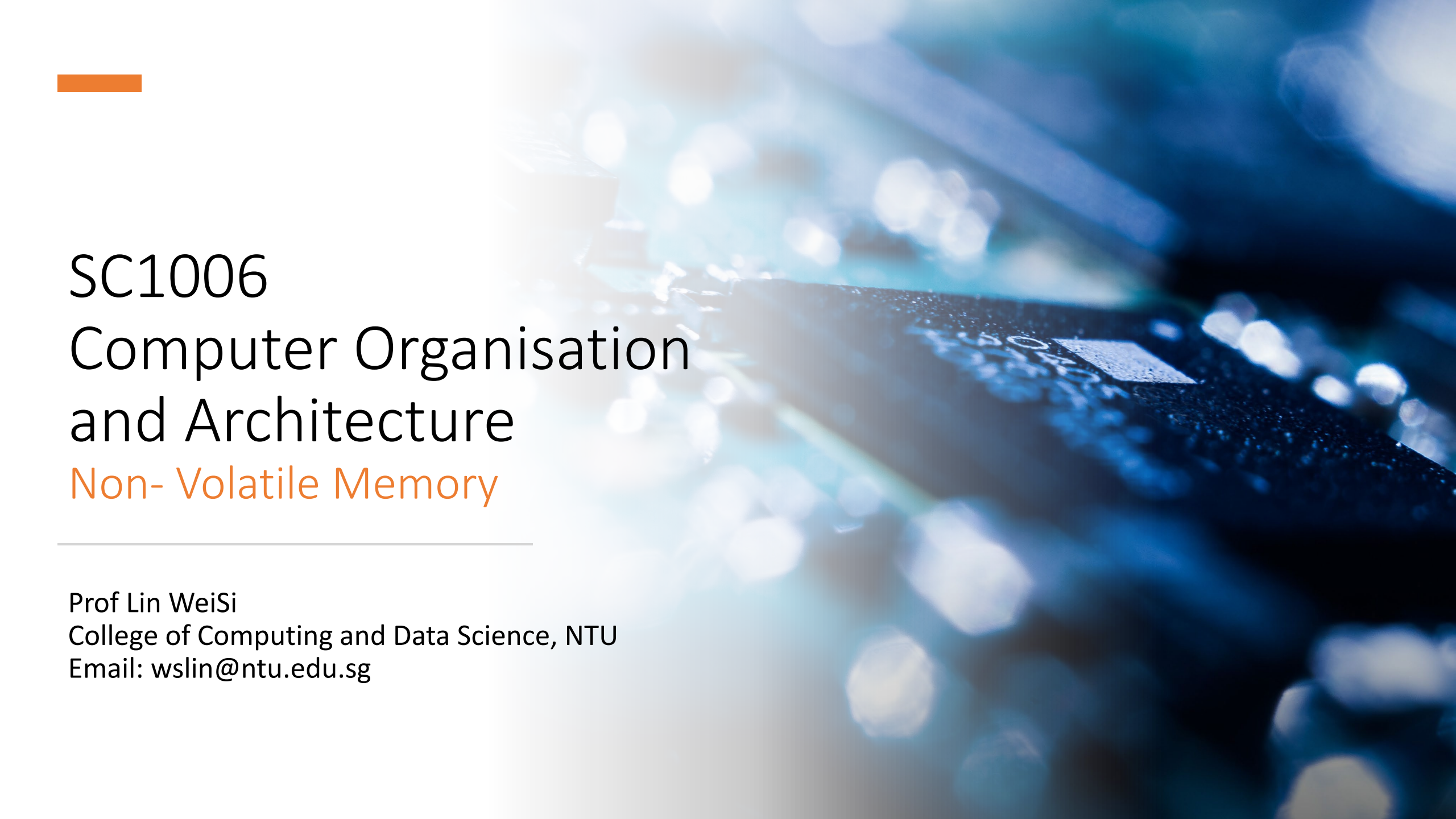




SC1006

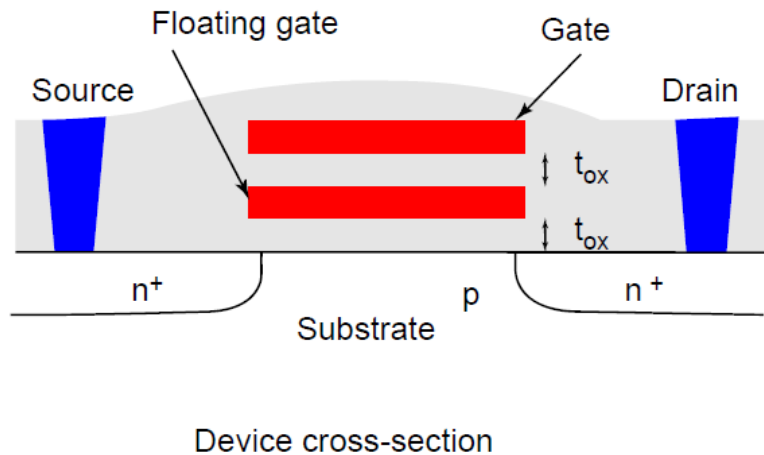
Computer Organisation and Architecture

Non- Volatile Memory

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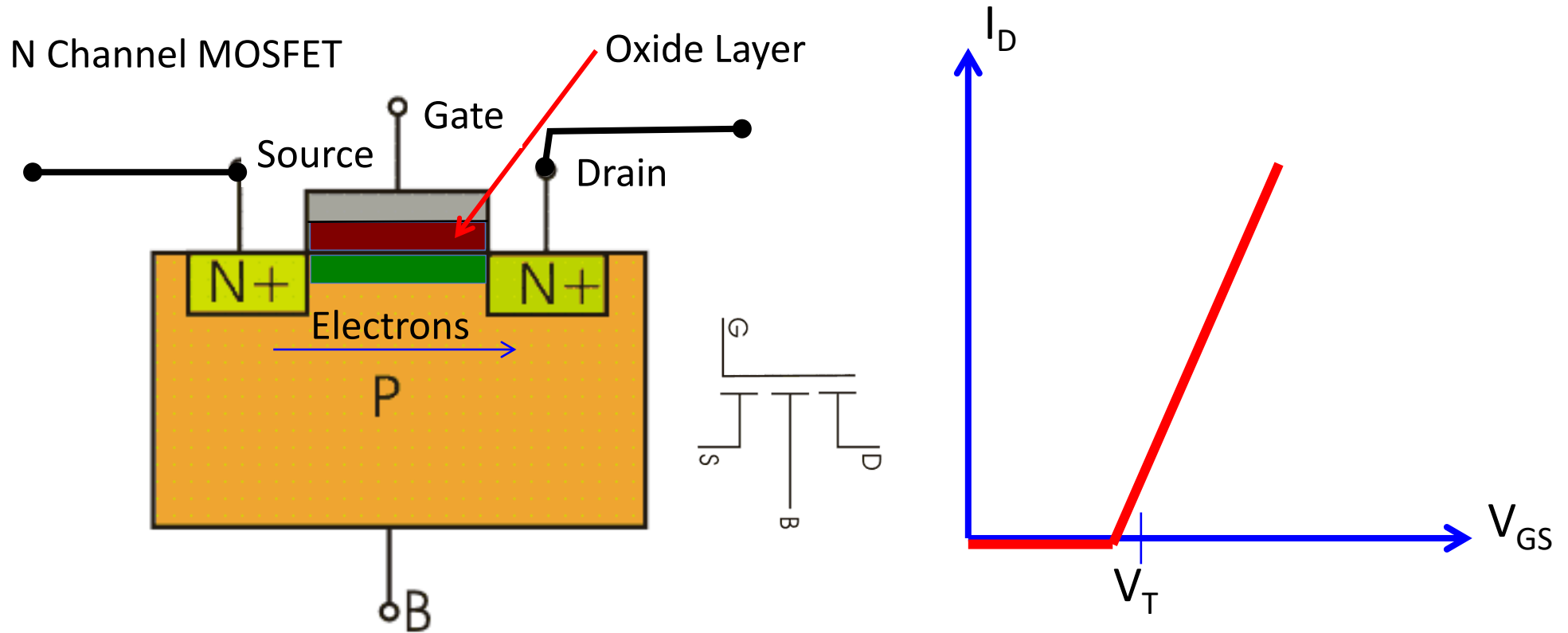
EPROM and EEPROM

- EPROM (Erasable Programmable ROM)
 - Earliest floating gate transistors are implemented as Erasable Programmable ROM (EPROM) devices.
 - Need to put device under ultra-violet (UV) light to **erase** the stored program.



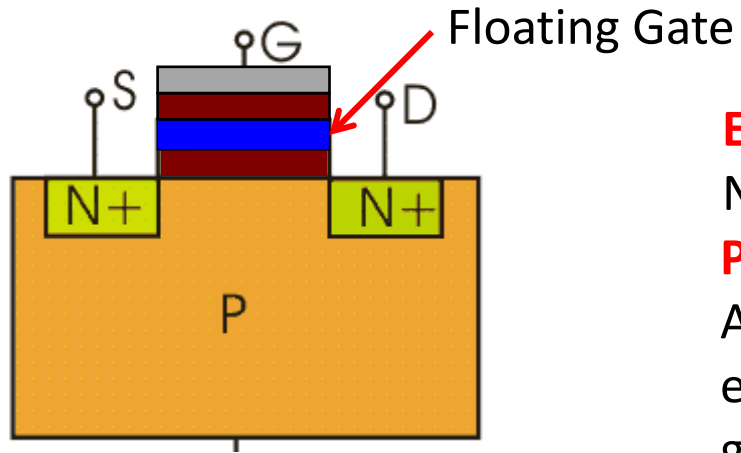
- EEPROM (Electrically Erasable PROM)
 - Advancement in process technologies makes it possible to **reduce** the **oxide thickness** (t_{ox}).
 - Can **electrically program or erase devices**.
 - Hence, named Electrically Erasable Programmable ROM (E²PROM).

A floating gate in MOSFET



- MOSFET (Metal-Oxide-Semiconductor Field Effect Transistor)
- For N-Channel MOSFET, if Gate-Source Voltage (V_{GS}) > Threshold Voltage (V_t), a **conductive** channel of electrons (inversion layer) will be formed and current will flow if a **positive voltage** is applied across Drain and Source.

Floating Gate Transistor (FGT)



Erased State

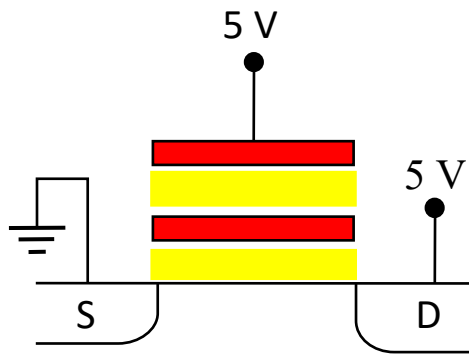
No Excess Electrons in Floating Gate

Programming

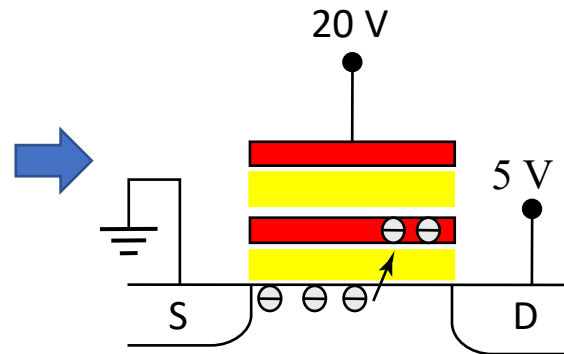
A large positive voltage applied at the gate, causing excess electrons to tunnel through the oxide and embed in the floating gate

Programmed State

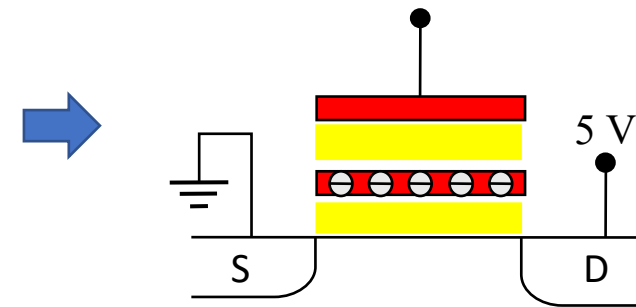
Excess Electrons are trapped in Floating Gate even after power off



Erased State



Programming



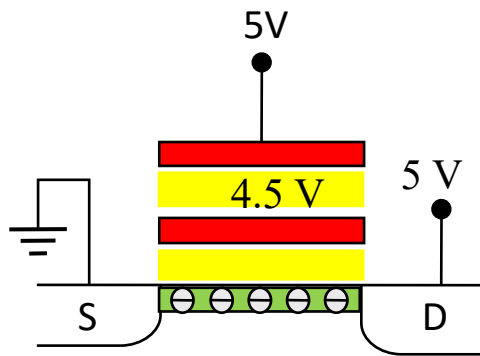
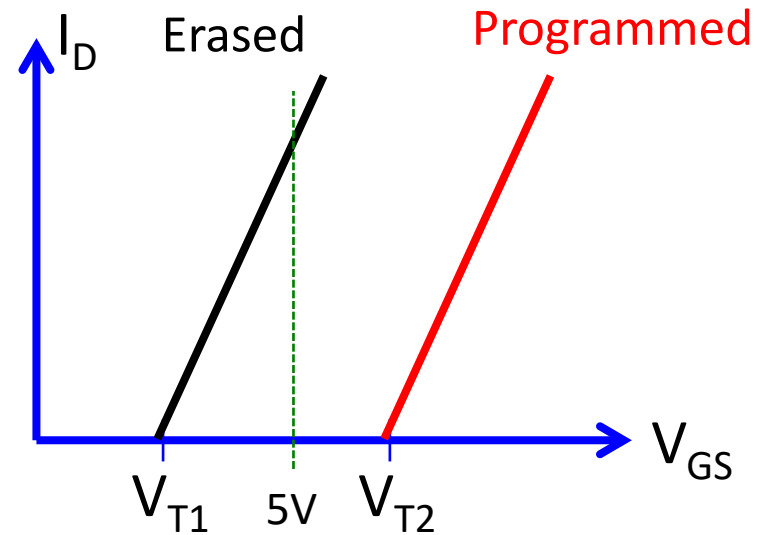
Programmed State

FGT Threshold Voltage

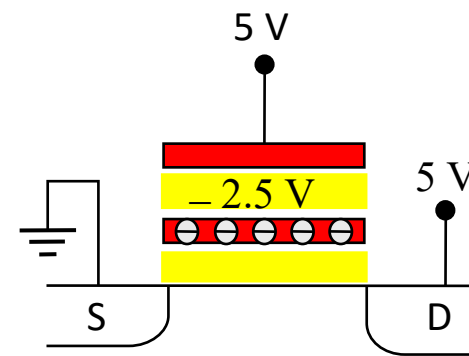
As an illustration, let's say a 5V input at the Gate of a FGT in erased state is sufficient to turn ON the FGT.

"Programming" increases the threshold voltage of Floating Gate Transistor so the same 5V is now unable to turn ON the FGT.

Voltage values used are for illustration purpose only.



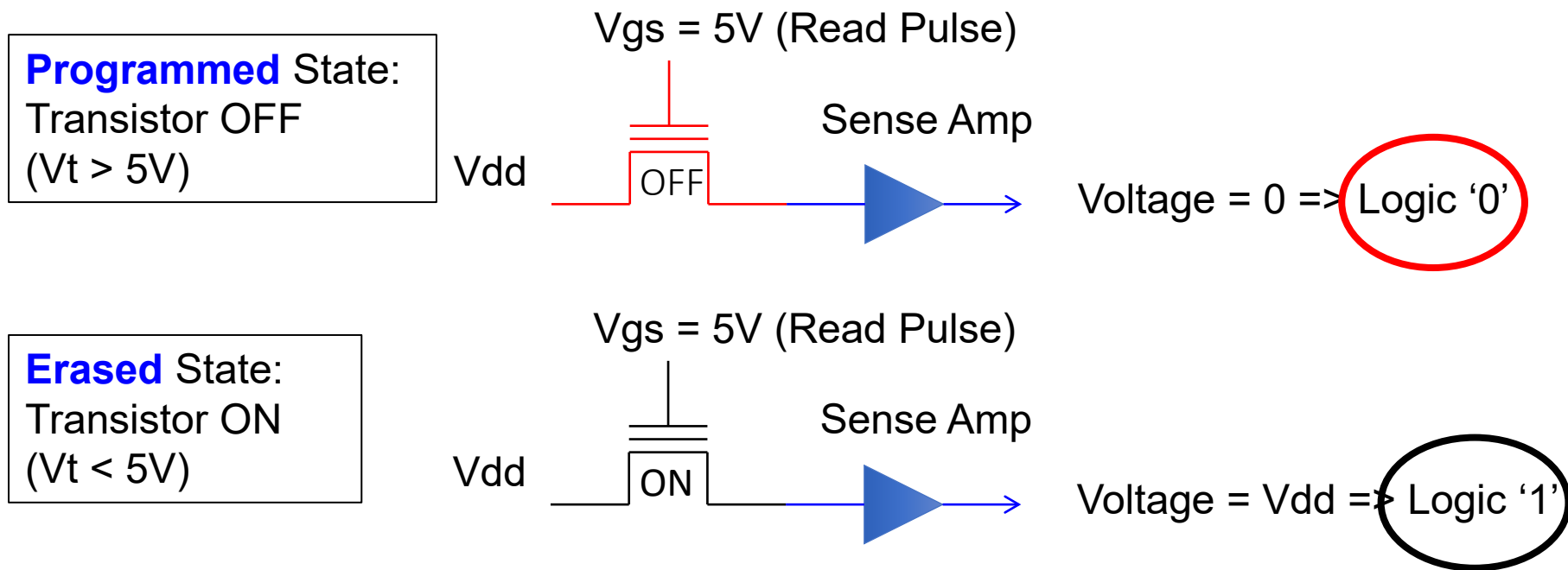
Erased State



Programmed State

Reading of Stored Data in Floating Gate

- With a positive Gate Voltage ($V_{gs} = 5V$)
 - Transistor **OFF** if floating gate is **programmed** (contains charges).
 - Transistor **ON** if floating gate is **erased** (no charges).
- 5V value below is for **illustration only**. Actual V_t value in real world may varies depending on the doping of the transistors.



Example of Programming a FGT based memory

- Flash '**programming**' can only modify the **cell content from '1' to '0'**
- To modify the **cell content from '0' to '1'**, an **erase operation** has to be done at block level
- Example: To Program a value of '**0x45**' when initial value in flash memory is '**0x55**'

Initial = 0x55

0	1	0	1	0	1	0	1
---	---	---	---	---	---	---	---

Block Erase

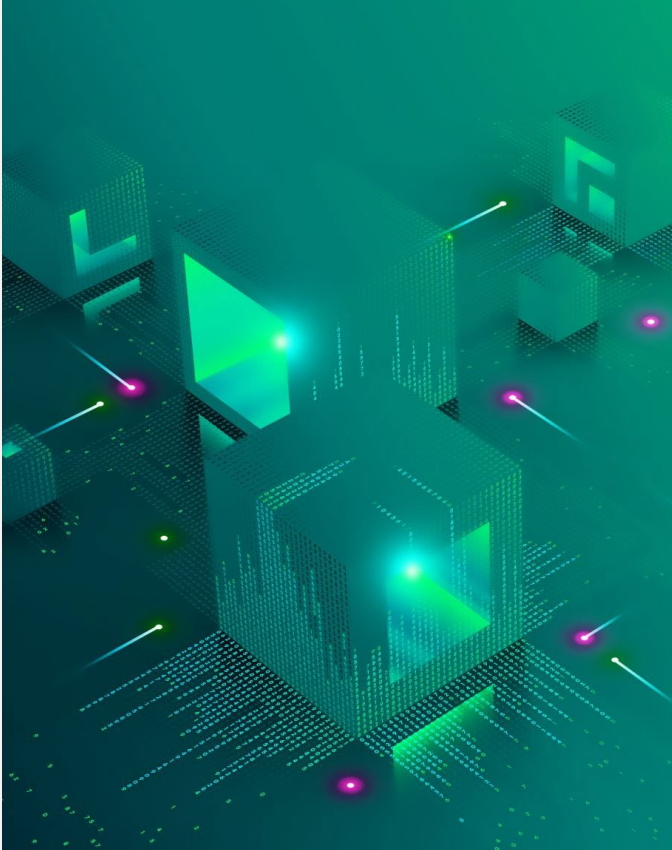
1	1	1	1	1	1	1	1
---	---	---	---	---	---	---	---

Programming involve 'pulling' the '1's to '0's

Final = 0x45

0	1	0	0	0	1	0	1
---	---	---	---	---	---	---	---

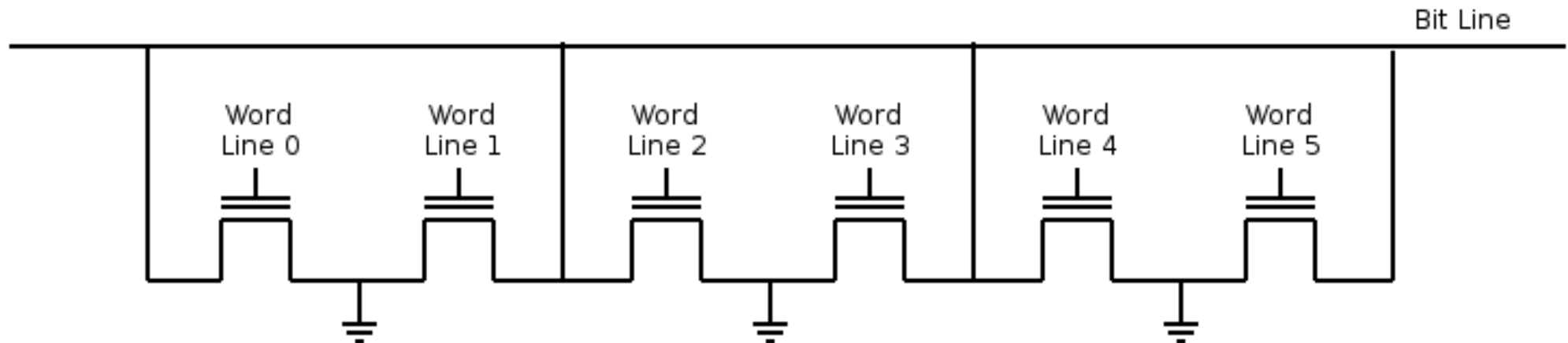
Flash



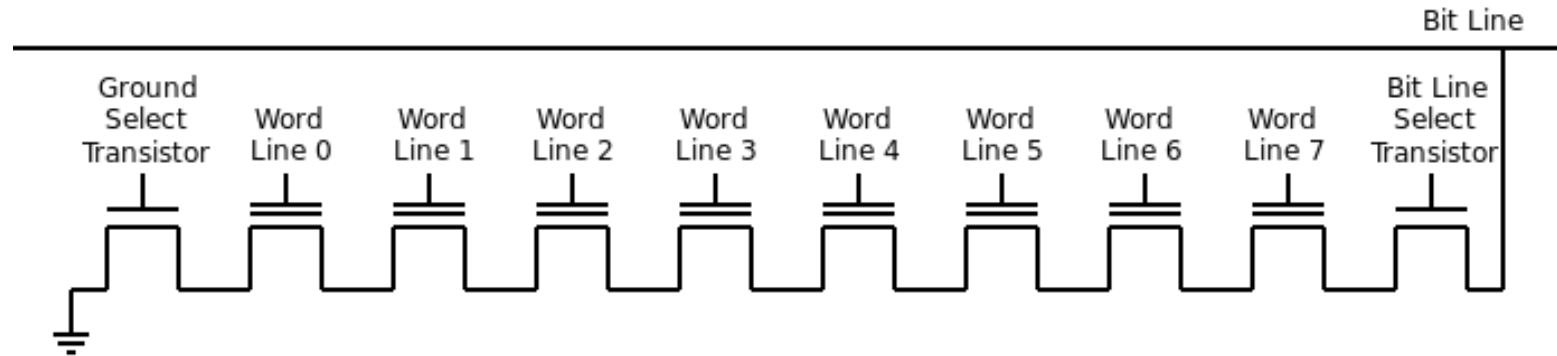
- Based on similar floating gate technology as EEPROM.
- Can be erased in larger blocks size compared to EEPROM (which typically has smaller page/block size).
- Flash memory has a faster speed than EEPROM when performing write operations for large block of data.
- Flash also cost less than EEPROM.
- Suitable for system requiring large amount of non-volatile memory.
- Two main types of Flash in the market
 - NAND Flash
 - NOR Flash

NOR Flash

- Cell connects like a NOR gate.
When any one of the Word Line $> V_t(\text{Prog})$, Bit Line output = 0.
- Supports Execute in Place, i.e. Program code stored in Parallel NOR Flash can be executed directly without the need to transfer to internal RAM first.
- Allows Random Reading of memory data using only Address information (no additional Commands needed).
- Need Special Commands (issue in Write mode) in order to perform operations other than Data Read. E.g. Program, Erase etc.
- Allows random word/byte programming. But erasure has to be done at block level. Typical Block size: 64KByte, 128KByte, 256KByte
- Use as system memory to store program code or general storage memory

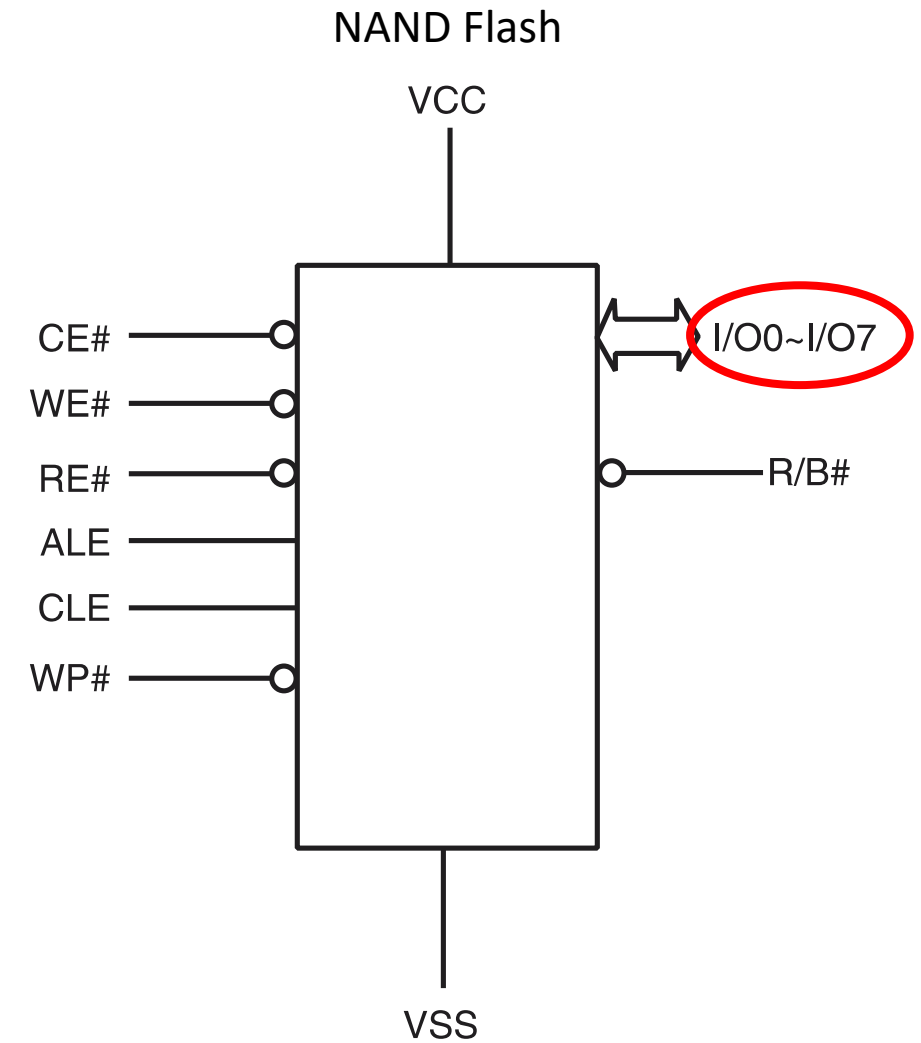
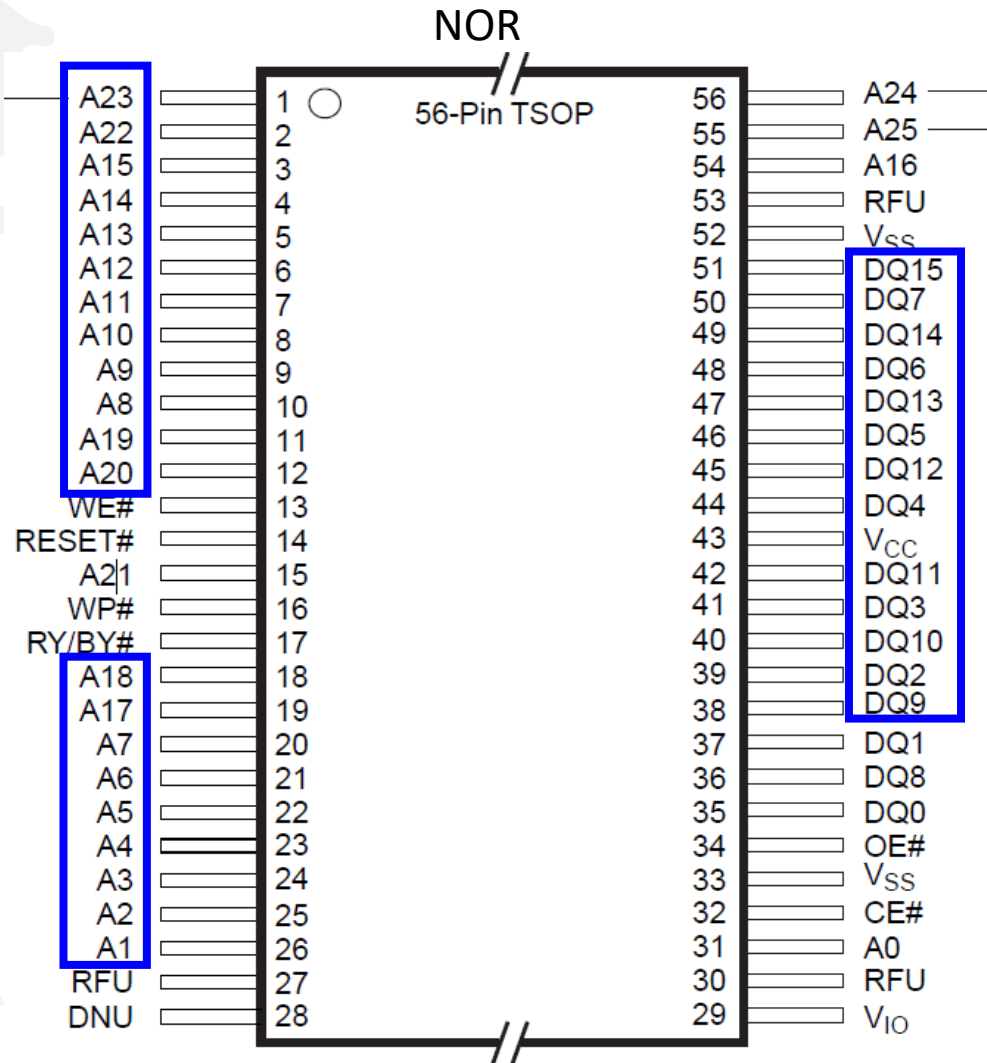


NAND Flash

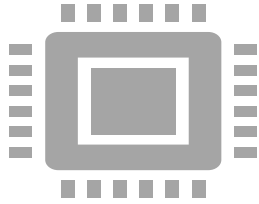


- Cell behaves like NAND gate.
 - Bit line output = '0' only when ALL Word Line $> V_t(\text{Prog})$.
- Does not support execute in place operation.
 - Data has to be accessed one page at a time.
 - Command issued to open a particular page, followed by which byte(s) is/are needed in the page.
 - NAND chips uses a single bus to carry Address and Data.
- Lower Cost per Bit than NOR. Used mainly as Main Storage Memory.
 - On Board Main Storage, USB Flash Drive, SSD etc

NOR and NAND Flash chip pin-out



System vs Storage Memory



System Memory

Used to store runtime program and code is executed directly from the system memory

This typically

- Use SRAM/DRAM or NOR Flash
- Support XIP

DRAM technically speaking does not support XIP by itself (as its access is multi-steps) but processors that has a DRAM interface will have DRAM controller that handle the necessary translation to allow execution of code directly from the DRAM.



Storage memory

Used to store all program and data in a computer system

Cannot run code directly from storage memory, needs to be transferred to system memory before code execution

All memory types can be used as storage memory

In practice, you usually need the storage memory to be able to retain information when the power is cut off, then **non-volatile memory** has to be used.

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